FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – V
(Paper-1)

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
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2.	**B**	20.	**D**	38.	**D**
3.	**C**	21.	**A**	39.	**D**
4.	**B**	22.	**B**	40.	**D**
5.	**B**	23.	**D**	41.	**C**
6.	**A, C, D**	24.	**A, B, D**	42.	**B, C, D**
7.	**A, C**	25.	**C**	43.	**B, C**
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14.	**4**	32.	**4**	50.	**9**
15.	**8**	33.	**5**	51.	**4**
16.	**4**	34.	**4**	52.	**3**
17.	**3**	35.	**0**	53.	**2**
18.	**8**	36.	**4**	54.	**2**

Physics

PART – I

SECTION – A

1. **A**

Acceleration slowly increases from zero to α, so small sphere will be at rest w.r.t ground when maximum acceleration α is to be reached in frame of cylinder, small sphere will have αr downward acceleration applying condition of rolling

$$\alpha r = \frac{g\sin\theta + \alpha r}{1+\frac{2}{5}}$$

$$\frac{7}{5}\alpha r - \alpha r = g\sin\theta$$

$$\theta = \sin^{-1}\left(\frac{2\alpha r}{5g}\right)$$

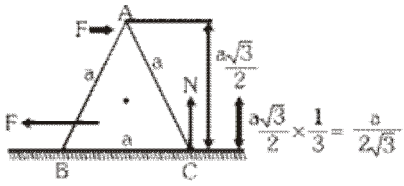
2. **B**

3. **C**

$$I = \frac{ma^2}{6} + m\left(\frac{a}{\sqrt{3}}\right)^2 \Rightarrow I = \left(\frac{ma^2}{2}\right)$$

$$\Rightarrow F \times \frac{a\sqrt{3}}{2} = (mg)\frac{a}{2} \Rightarrow F = \left(\frac{mg}{\sqrt{3}}\right)$$

$$F = F_{max} \Rightarrow \frac{mg}{\sqrt{3}}\mu mg \Rightarrow \mu = \frac{1}{\sqrt{3}}$$



4. **B**

$$y_{cm} = \frac{\sum m_i y_i}{\sum m_i} \text{ and } x_{cm} = \frac{\sum m_i x_i}{\sum m_i}$$

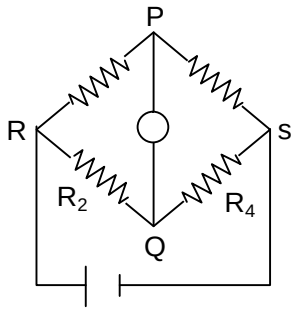
5. **B**

Momentum & energy conservation $6 \times 20 = (24+6)v$

v = 4 m/s.

6. **A, C, D**

Equivalent diagram is as shown is P is moved 2 cm right them $R_1 = 12, R_3 = 3$ $\frac{R_1}{R_3} = \frac{R_2}{R_4}$ (Hence wheat stone will be balanced) If s is moved left $\frac{5}{3}$ cm then $R_3 = \frac{10}{3}$ and $R_4 = \frac{20}{3}$ hence $\frac{R_1}{R_3} = \frac{R_2}{R_4}$ (hence Wheatstone will be balanced)



7. **A, C**

R is independent of freq,

(A) is true.

Current becomes maximum at resonance

(B) cannot be true

$X_L = L = 2\pi fL$

(C) is true

$Y = \frac{1}{C} = \frac{1}{2\pi fC}$

(D) can not be true.

8. **A, C**

Asteriod – I

Field $E = \frac{GM}{R^3}x$

$\frac{F_I}{F_{II}} = \frac{2x}{R}$

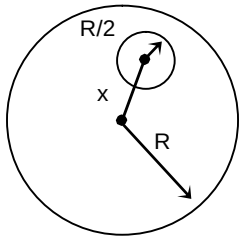
$T_I = \frac{2\pi}{4}\sqrt{\frac{R^3}{GM}}$

Asteriod – II

Field $E_{II} = \frac{GM}{R^3}\left(\frac{R}{2}\right)$

$R = \frac{1}{2}\frac{GM}{R^3}\left(\frac{R}{2}\right)T_{II}^2$

$T_{II} = 2\sqrt{\frac{R^3}{GM}}$



9. **A, B, C, D**

$\vec{E} = (10 - 5x)i$

$m = 5kg$

$\vec{F} = q\vec{E}$

$\vec{F} = (10 - 5x)i$

$(a)\ F \propto -x$

so motion of charge particle is oscillatory (b) F = 0 at x = 2

oscillation about x = 2

so maximum displacement is 4

$F = 10 - 5x$

$a = \frac{10 - 5x}{2}$

$a = 2 - x$

$a = -(x - 2)$

so $w^2 = 1$

$w = 1$

$(c)\ v_{max} = Aw$

$A = 2$

$w = 1$

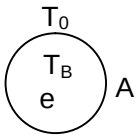
$v_{max} = 2$

(d) velocity at mean position at x = 2

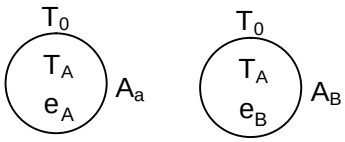
10. **A, C, D**

11. **A, D**

12. **A, B, C, D**



Emissivity of body e
Area of body A
Temperature of body T_B
Temperature of environment T_0
Heat energy radiated by body $= e\sigma AT_B^4$
Absorbed by body = $e\sigma AT_0^4$

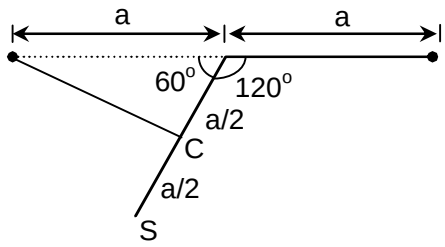


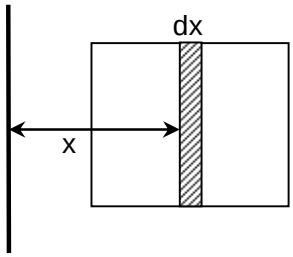
$e_A \neq e_B$
Then heat emitted by heat emitted by a body is not equal to B.

13. **B, C, D**
The probability of decay in 1 sec is the decay constant D, which remains constant after average life 37% of nuclei remain undecayed part of energy of disintegration is taken by the recoiling daughter nucleus.

SECTION – C

14. **4**
After moving the loop about side PQ through 120°, the normal on the side PS from the wire will be passing through midpoint C and the net magnetic flux through the loop in the new position will be zero.
$\therefore$ Changes is flux $\Delta f =$ Flux through the loop in the initial position.
Flux through area element dA





$$d\phi = \vec{B}.d\vec{A}\frac{\mu_0 I}{2\pi x}a.dx$$

$$\phi = \int_a^{2a}\frac{\mu_0 I a dx}{2\pi x} = \frac{\mu_0 I a}{2\pi}\ell n(2)$$

$\therefore$ Charge flown through the square loop $= \frac{\Delta\phi}{R} = \frac{\mu_0 I a}{2\pi R}\ell n(2)$

15. **8**

Height of water column $= \frac{20}{5} = 4$ cm

Mass of water = 20 gm

By work energy theorem

$W_{ext} = \Delta U_{piston} + \Delta U_{water} + KE_{water} = 8 \times 10^{-2}$ Joule

16. **4**

$W_g + W_{mg} + W_{friction} = \Delta KE$

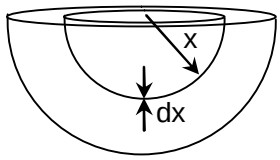
$W_{friction} = 0 - \frac{1}{2} m \left[\frac{qBR}{m}\right]^2$

$W_{friction} = -\frac{1}{2}\frac{q^2B^2R^2}{m} = \frac{-Kq^2B^2R^2}{8m}$

K = 4

17. **3**

$V = \int_a^b \frac{K\left(2\pi x^2 dx\right)\rho}{x}$



$V = 2\pi k\rho\left[\frac{b^2 - a^2}{2}\right] = \pi\rho k\left(b^2 - a^2\right)$

$b = 2a$

$v = 3\pi k\rho b^2$

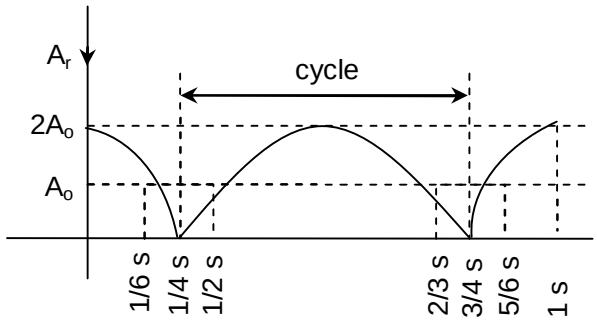
18. **8**

$y_1 = A\sin\omega_1 t$

$y_2 = A\sin\omega_2 t$

$y_1 = 2A\cos\left\{\frac{(\omega_2 - \omega_1)}{2}\right\}\left\{\sin\frac{(\omega_2 + \omega_1)}{2}t\right\}$

Resultant amplitude $A_r = 2A_0|\cos(\Delta\omega)t/2|$



$(\Delta\omega)\frac{t}{2} = \frac{\pi}{2} \Rightarrow t = \frac{1}{4}$s

$(\Delta\omega)\frac{t}{2} = \frac{\pi}{3} \Rightarrow t = \frac{1}{6}$s

In one cycle of intensity of ½ s, the detector remain idle for

$2\left(\frac{1}{4} - \frac{1}{6}\right)s = \frac{1}{6}$sec

$\therefore$ In 1/2 sec cyclem active time is $\left(\frac{1}{2} - \frac{1}{6}\right)$

$= 1/3$ sec

$\therefore$ In 12 sec interval, active time is $12 \times \frac{(1/3)}{(1/2)} = 8$ sec

Chemistry

PART – II

SECTION – A

19. **C**

After adding water equilibrium will shift backward and new concentration will be less than $\frac{x}{2}$.

20. **D**

A is $Na_2S_2O_3$, & C is Ag_2S.

21. **A**

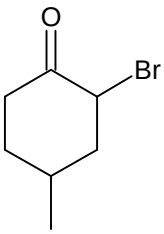
$r_1 = \lambda N_1$

$8100 = \lambda N_o$

$2700 = \lambda N_t$

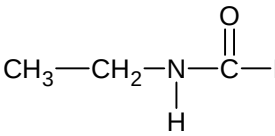
$\lambda = \frac{2.303}{50}\log\frac{N_o}{N_t} = \frac{0.693}{t_{1/2}}$

22. **B**



= A, A ⟶ B favorskii rearrangement

23. **D**

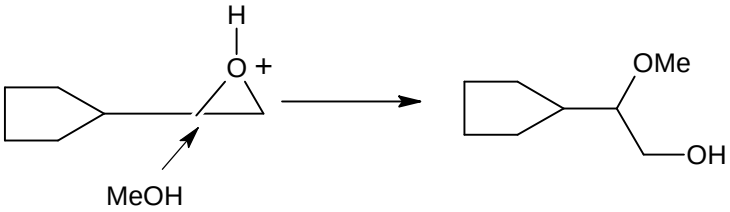


has acid amide functional gp

24. **A, B, D**

Hemiacetals give Tollen's test fructose $(\alpha - \text{hydroxy ketone})$ also gives Tollen's test.

25. **C**



26. **A, B, C**

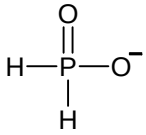
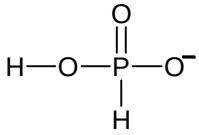
In Cerric ammonium nitrate test alcohol produce red color due to formation of $(NH_4)_2\left[Ce(OR)(NO_3)_5\right]$ while phenol produces brown.

27. **A, B, C**

$\left[Pt(NH_3)_4Cl_2\right]Cl_2$, $\left[Ni(en)_3\right]Cl_2$ both have same number of Cl^- in ionization sphere, Zeise's salt is $K\left[PtCl_3(C_2H_4)\right]$, $K_4\left[Fe(CN)_6\right]$ & $K_3\left[Fe(CN)_6\right]$ both have d^2sp^3.

28. **C, D**

Cl is most electronegative among Cl, S, P & C so ClO_2 is most acidic and Ea of O is least in its

gp ;  , 

1 2

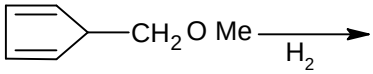
Due to cross conjugation in (2) ; 1 is weaker base than 2 so H_3PO_2 is stronger acid than H_3PO_3, similarly relate H_3PO_4 for option (b) refer e^- configuration & ENC $\Rightarrow$ Na > Al > Si > Mg.

29. **A, B**

XeF_2 & XeF_4 dissolve in liq HF without any chemical reaction while XeF_6 reacts with HF & $NaH + HF \longrightarrow H_2 + NaF$.

30. **A, B, D**

A is  Li ; B is C_4H_{10}

C is 

has highest density among R - X

31. **A, B, C, D**

$IE_2 > IE_1$ (as ENC increases)

$O^- + e^- \longrightarrow O^{2-}$

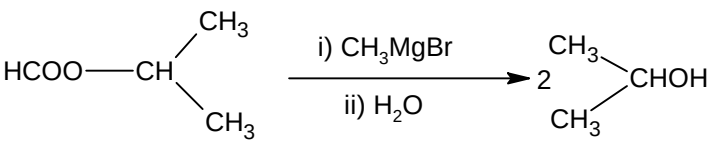
Due to FOR between O^- and e^- the process is endothermic.

F has highest electronegativity.

First I.P. of N is less than first I.P. of F.

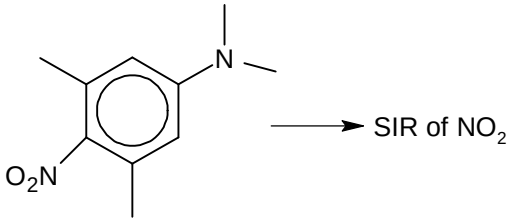
SECTION – C

32. **4**



33. **5**

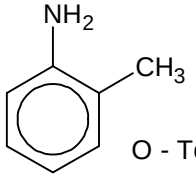
$CH_3 - Cl > CH_3 - F(\mu = q \times d)$



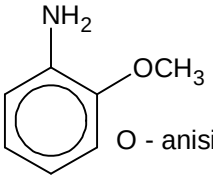
$CH_2 = CHCl \Rightarrow$ –I is opposed by + R of Cl

34. **4**

Refer ortho effect and resonance, inductive effect etc.



O - Toluidine



O - anisidine - ortho effect does not work.

35. **0**

Acetylene gives Tollen's test but not form silver mirror (Ag). (Ag_2C_2 - white ppt.)

36. **4**

Refer FCC.

Mathematics

PART – III

SECTION – A

37. **B**

$\because 2\cos\theta_1 = \frac{8}{5} \Rightarrow \cos\theta_1 = \frac{4}{5} \Rightarrow \tan\frac{\theta_1}{2} = \sqrt{\frac{1-\frac{4}{5}}{1+\frac{4}{5}}} = \frac{1}{3}$ and $e = \frac{1}{2}$

$\therefore \tan\frac{\theta_2}{2} \times \frac{1}{3} = \frac{-\frac{1}{2}}{\frac{3}{2}}$

$\frac{\theta_2}{2} = \frac{3\pi}{4} \Rightarrow \theta_2 = \frac{3\pi}{2}$

$\therefore B = \left(0,\ -\sqrt{3}\right) \Rightarrow AB = 2 + 2 - \frac{1}{2} \times \frac{8}{5} = 4 - \frac{4}{5} = \frac{16}{5}$

38. **D**

$\because$ Image of (2, 1) in $2x - y + 1 = 0$ is

$\frac{x-2}{2} = \frac{y-1}{-1} = \frac{-2(4-1+1)}{4+1} = \frac{-8}{5}$

$\Rightarrow x = 2 - \frac{16}{5} = \frac{-6}{5}$

$y = \frac{8}{5} + 1 = \frac{13}{5}$

$\therefore$ Slope of directrix $= \frac{\frac{13}{5} - 3}{\frac{-6}{5} - 1} = \frac{-\frac{2}{5}}{-\frac{11}{5}} = \frac{2}{11}$

$\therefore$ Equation of axis,

$y - 1 = \frac{-11}{2}(x - 2)$

$\Rightarrow 2y - 2 = -11x + 22$

$\Rightarrow 11x + 2y = 24$

39. **D**

$\because \left(\hat{a} \times \hat{b}\right).\left\{(\hat{a}.\hat{c})\hat{b} - (\hat{a}.\hat{b})\hat{c}\right\} = 0 - (\hat{a}.\hat{b})\left[\hat{a}\ \ \hat{b}\ \ \hat{c}\right]$

$= -\frac{1}{2} \times \frac{1}{\sqrt{2}}$

$\therefore \left[\hat{a}\ \ \hat{b}\ \ \hat{c}\right]^2 = \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix} = \frac{1}{2}$

40. **D**

$\because a_1^2+b_1^2+c_1^2+a_2^2+b_2^2+c_2^2+a_3^2+b_3^2+c_3^2=7$, where $\langle a_i,\ b_i,\ c_i\rangle$ are elements of ith row.

$\therefore$ Case I: All seven entries are 1 and rest of all zero

${}^9C_7 = {}^9C_2 = 36$

Case II: Only one entry is 2 & three entries are 1 and rest of all 0's.

${}^9C_4 \times {}^4C_1 = 504$

$\therefore$ Total = 540

41. **C**

$$\because \frac{1}{4}\sin 3\theta = \frac{3}{4}\sin\left(\frac{\theta}{9}\right) - \sin^3\theta - 3\sin^3\frac{\theta}{3} - 9\sin^3\left(\frac{\theta}{9}\right)$$

$$\Rightarrow \sin 3\theta = \frac{1}{\sqrt{2}} = \sin\frac{\pi}{4} \text{ or } \sin\frac{3\pi}{4}$$

$$\Rightarrow \theta = \frac{\pi}{12} \text{ or } \frac{\pi}{4}$$

$$\therefore \tan\frac{\pi}{12} = 2-\sqrt{3} \text{ or } \tan\frac{\pi}{4} = 1$$

$$\therefore \tan\frac{\pi}{12} + \tan\frac{\pi}{4} = 3-\sqrt{3}$$

42. **B, C, D**

$$\because \left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right] = 2 \text{ and } 3\left(\bar{a}+\bar{b}\right).\left\{\left(\bar{b}+\bar{c}\right)\times\left(\bar{c}+\bar{a}\right)\right\}$$

$$= 3\left(\bar{a}+\bar{b}\right).\left\{\bar{b}\times\bar{c}+\bar{b}\times\bar{a}+0+\bar{c}\times\bar{a}\right\}$$

$$= 3\left\{2\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]\right\} = 6\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right] = 12$$

$$\left[\bar{a}\times\bar{b}\ \ \bar{b}\times\bar{c}\ \ \bar{c}\times\bar{a}\right] = \left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]^2 = 4$$

$$\Rightarrow \left(\bar{a}-2\bar{b}\right).\left\{\left(\bar{b}-2\bar{c}\right)\times\left(\bar{c}-2\bar{a}\right)\right\} = \left(\bar{a}-2b\right)\left\{\bar{b}\times\bar{c}-2\bar{b}\times\bar{a}+4\bar{c}\times\bar{a}\right\}$$

$$= \left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right] - 8\left[\bar{b}\ \ \bar{c}\ \ \bar{a}\right] = -7\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]$$

$$\left(3\bar{b}-\bar{c}\right)\times\left(3\bar{c}+\bar{a}\right) = 9\bar{b}\times\bar{c}-3\bar{b}\times\bar{a}+\bar{c}\times\bar{a}$$

$$\left(3\bar{a}-\bar{b}\right).\left\{3\bar{c}+\bar{a}\right\} = 27\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right] - \left[\bar{b}\ \ \bar{c}\ \ \bar{a}\right] = 26\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]$$

43. **B, C**

$\because \sqrt{6}x\sec\theta - \sqrt{3}y\,\text{cosec}\,\theta = 3$ is equation of normal,

then, slope $= \dfrac{-\sqrt{6}\sec\theta}{-\sqrt{3}\,\text{cosec}\,\theta} = \sqrt{2}\tan\theta$

Given, $\sqrt{2}\tan\theta = +1$

$$\Rightarrow \tan\theta = \frac{1}{\sqrt{2}}$$

$\therefore$ normal is $x-y=1$

Let $P\left(h,\ \dfrac{2}{h}\right)$

$\Rightarrow h-\frac{2}{h}=1$

$\Rightarrow h^2-h-2=0$

$\Rightarrow h=2 \text{ or } -1$

44. **A, B, D**

$\because y=mx-2m-m^3$

$\Rightarrow 6=9m-2m-m^3$

$\Rightarrow m^3-7m+6=0 \Rightarrow m=1,+2,-3$

$\therefore$ Centre is $\left(\frac{11}{2},\ \frac{3}{2}\right)$

45. **A, B, C**

$\because f(x)=2\times\max\left\{|x-1|+|x-3|,\ |2x-1|\right\}$

$\therefore x=1,\ \frac{3}{2}$ point of non-differentiability and minimum value is 2

46. **A, B, D**

$\because \int\cos(2017x+x)\sin^{2016}x\,dx=\int\cos 2017x\sin^{2016}x\cos x\,dx-\int\sin 2017x\sin^{2017}x\,dx$ ..... (i)

$\left\{\because \int\cos\underset{(I)}{2017x}\underbrace{\sin^{2016}x\cos x}_{(II)}\,dx=\frac{\cos 2017x\sin^{2017}x}{2017}+\int\frac{2017\sin 2017x}{2017}\sin^{2017}x\,dx\right\}$

$\therefore \int\cos 2018x\sin^{2017}x\,dx=\frac{1}{2017}\cos 2017x\sin^{2017}x$

47. **A, B, C**

$\because a^2-(b+c)^2+xbc=0$

$\therefore x=\frac{(a-b-c)(a+b+c)}{bc}$

$=\frac{4s(s-a)}{bc}=4\cos^2\frac{A}{2}\in(0,\ 4)$

$\because 0<A<\pi$

$\Rightarrow 0<\frac{A}{2}<\frac{\pi}{2}$

48. **A, B, C**

$\because f'(x)=b^2+(a-1)b+2-(\sin^2x+\cos^4x)\geq 0$

$(f'(x))_{min}=b^2+(a-1)b+1\geq 0$

$\therefore (a-1)^2-4<0\Rightarrow a\in(-1,\ 3)$

49. **B, D**

$\because f(x)=\frac{xf'(x)}{2\sqrt{f(x)}}+\sqrt{f(x)}$

$\Rightarrow \sqrt{f(x)} = \dfrac{1}{1-cx}$

$\Rightarrow f(x) = \dfrac{1}{(1-cx)^2}$

$\because f(1) = \dfrac{1}{2} \quad \therefore c = 1-\sqrt{2}$

$\Rightarrow f(x) = \dfrac{1}{\left[1+\left(\sqrt{2}-1\right)x\right]^2}$

SECTION – C

50. **9**

$\dfrac{x-2}{-\dfrac{1}{\sqrt{2}}} = \dfrac{y-1}{\dfrac{1}{\sqrt{2}}} = 4\sqrt{2}$ or $-4\sqrt{2}$

$\therefore A(-2,\ 5)$ & $B(6,\ -3)$

$\therefore$ area of $\Delta CAB = 20$

$\Rightarrow$ sum $= 2\sqrt{20} = 4\sqrt{5}$

51. **4**

$\sqrt{2\left(a^2+b^2\right)} = 4 \Rightarrow a^2+b^2 = 8$

$\therefore \dfrac{a^2+b^2}{2} \geq ab$

$\Rightarrow ab \leq 4$

52. **3**

$\because x^2+5x-6 = 0$

$\Rightarrow (x-1)(x+6) = 0 \Rightarrow x = 1$

$\therefore x+5y = 6$ ..... (i) tangents

& $y = \dfrac{1}{2}(x+1)$ ..... (ii)

$\therefore \tan\theta = \left|\dfrac{-\dfrac{1}{5}+\dfrac{1}{2}}{1-\dfrac{1}{10}}\right| = \dfrac{3}{9} = \dfrac{1}{3}$

53. **2**

$\dfrac{a+b}{2} = \dfrac{5}{4}\sqrt{ab}$

$\Rightarrow \dfrac{a}{b} = \dfrac{1}{4}$

$\dfrac{H_8-a}{a-H_8} = 8\times\dfrac{a}{b} = 8\times\dfrac{1}{4} = 2$

54. **2**

$$\because f(x)=\sum_{i=1}^{\infty}\left[\frac{1}{(i-1)x+1}-\frac{1}{ix+1}\right]$$

$$\Rightarrow f(x)=\lim_{n\to\infty}\left[1-\frac{1}{nx+1}\right]$$

$$\Rightarrow f(x)=1$$

$$\therefore f\left(\frac{1}{x}\right)=1$$

$$\Rightarrow f(x)+f\left(\frac{1}{x}\right)=2$$